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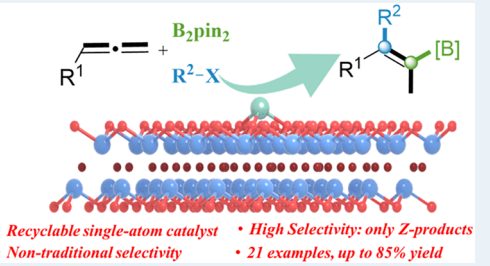
Regio- and Stereoselectivities for Alkylation and Boronization of Allenes through Metal−Support Electron Interactions of Single-Atom Catalysts

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Cite This: *ACS Catal.* 2025, 15, 15902−15909 Read Online

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ABSTRACT: In synthetic chemistry, controlling the regio- and stereoselectivities, especially nontraditional selectivities, of multicomponent tandem reactions has always been challenging. In organic reactions, single-atom catalysts (SACs) possess fixed coordination structures and adjustable metal−support interactions and frequently generate products possessing well-defined regio- and stereoselectivities. Herein, a silicon carbide (SiC)-supported copper SAC ($CuCl_2$@O−SiC) was synthesized using increased copper content and the interaction between the metal support and SiC was regulated through oxygen-based surface modification. Allenes underwent nontraditional β,γ-addition, exclusively generating *Z*-configured alkylboronization products. These findings revealed the β,γ-selectivity pattern in allene addition reactions, which advances the application of SACs in chemical catalysis. This work offers the potential for further developments in selective synthesis.



KEYWORDS: *regio- and stereoselectivities addition, allenes, carboboration, silicon carbide support, β,γ-addition, single-atom catalysts*

INTRODUCTION

Even for experienced synthetic chemists, rapid, efficient, and cost-effective synthesis of structurally simple multisubstituted olefins remains challenging;[1−4] however, multicomponent synthesis holds tremendous promise for synthesizing multisubstituted olefins.[5−7] Nevertheless, the simultaneous integration of diverse substrates within a single step inevitably amplifies the risk of side reactions and exacerbates the complexity of controlling olefinic stereochemistry.[8−13] Because of their distinctive dual perpendicular π-orbital structures and inherent reactivities, allenes have attracted considerable interest in organic synthesis.[14−19] Presently, precisely controlled α,β-additions of allenes through homogeneous copper/ligand catalytic systems are invaluable for directly fabricating sophisticated multisubstituted olefinic frameworks.[20−23] Among these reactions, the difunctionalization α,β-addition of allenes, at α-sp^2- and β-sp-hybridized carbon atoms, catalyzed by nucleophilic Cu−B species and predicated on borohydride reaction mechanisms, has emerged as a fervently discussed topic in synthetic chemistry (Figure 1A).[24−27] Despite recent advancements enabling the synthesis of various β,γ-addition products via α,β-addition rearrangements, copper predominantly metallates with α-carbon atoms.[28] Consequently, the multicomponent boron functionalization of allenes, generating branched products via β,γ-addition, remains exceptionally rare.

Navigating these challenges, the generation of internal alkynes from allenes in situ represents a promising approach for atypical selective additions.[29] However, this requires the simultaneous conquering of three pivotal hurdles: First, the conversion from allenes to alkynes must be expedited to suppress unnecessary α,β-addition of allenes to copper−boron intermediates. Second, during rearrangement and addition, the activity of copper catalysts must be sustained to prevent catalyst aggregation and deactivation for the completion of a single catalytic cycle. Finally, to finely tune the reaction's stereoselectivity, copper catalysts possessing befitting electronic effects must be developed.[30] In such intricate reactions, the catalyst must delicately balance sustained coordination stability and demanding chemo- and stereoselective regulations, which is difficult to achieve using traditional homogeneous metal/ligand catalytic systems (Figure 1B).[31]

In organic synthesis, the development of single-atom catalysts (SACs) is an important advancement in catalyst design.[32−42] Over the past decade, SACs have demonstrated remarkable chemical selectivities and superior catalytic

Received: June 10, 2025
Revised: August 22, 2025
Accepted: August 25, 2025
Published: August 30, 2025



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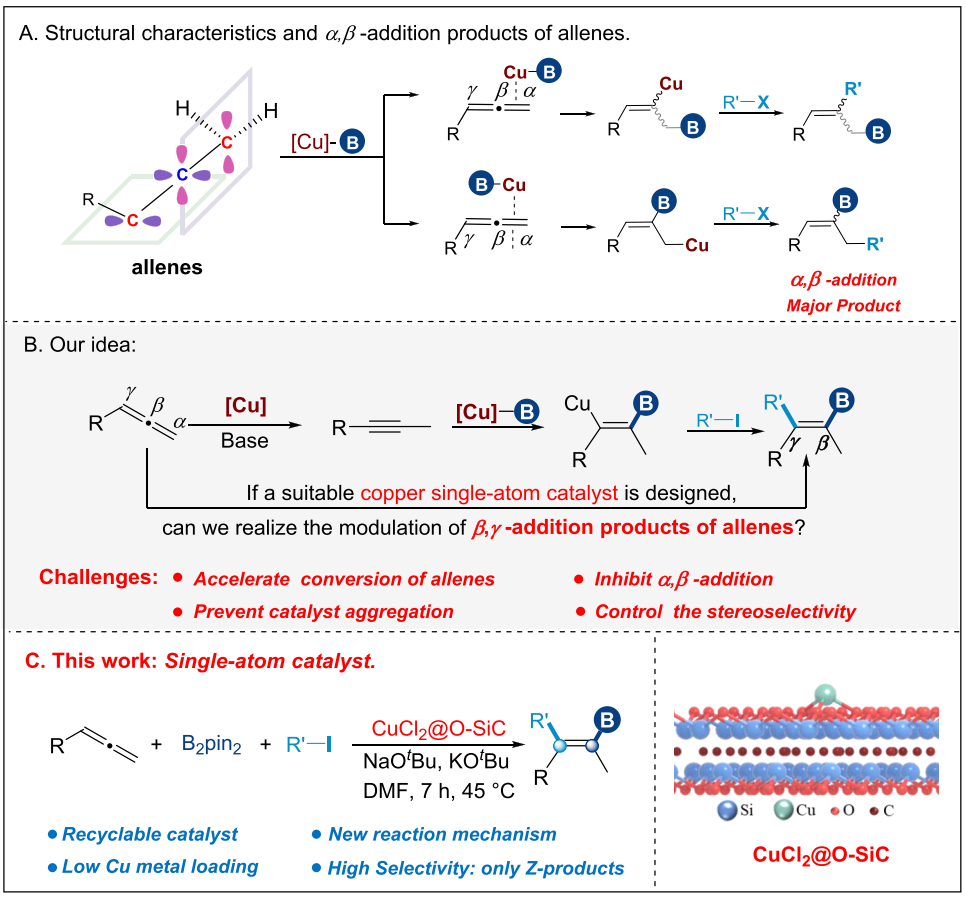


Figure 1. Metal-catalyzed selective alkylation and boronization of allenes.

activities because of their unique coordination structures and diverse fundamental characteristics derived from heterogeneous catalytic systems.[43−53] Chemists have sought to deeply understand the structure−catalytic activity dynamics of SACs and the manner in which the carrier regulates the electronic effects of metal catalytically active centers. Consequently, the mechanisms by which SACs coordinate and determine the stereoselectivities of organic reactions have gradually been elucidated.[54] SACs' distinct inherent coordination abilities make them suitable for satisfying conflicting chemical selectivity requirements, and their robust coordination structures and remarkable catalytic activities prevent catalyst aggregation during tandem reactions, thus substantially improving the catalytic efficiency. By modulating the electronic interplay between the metal and support, SACs may enable the control of the stereoselective addition of Cu−B intermediates, offering an alternative approach to form β,γ-addition products. (Figure 1C).

METHODS

To demonstrate the catalyst's efficacy, a series of experiments were conducted using a homogeneous catalytic system; however, the results were unsatisfactory (Table S1, entries 1−8). To determine the effects of various transition-metal carbides on the selectivity and catalytic activity of alkyne hydroboration, a series of metal-carbide-supported Cu-based SACs was synthesized to evaluate their catalytic activities. The experimental results revealed that during the reaction, different metal-carbide supports exhibited distinct selectivities,[55] suggesting that the interaction between the metal support and catalyst plays a crucial role in modulating the regioselectivity of the reaction. Evidently, although both the chromium carbide (CrC) and titanium carbide (TiC) catalysts exhibited low selectivities, their different dispersibilities in the solvent hindered sufficient contact with the reaction substrates, considerably constraining their catalytic activities (Table S1, entries 9 and 10). After 72 h, the SiC catalyst carrier produced only a relatively low reaction yield of 50% (Table S1, entry 11). To investigate this peculiarity, the catalyst was characterized using inductively coupled plasma−optical emission spectroscopy (ICP−OES). The experimental results revealed that the catalyst's Cu content was as low as 0.011 wt % because of the high stability of the SiC support, which prevented surface oxidation, and the absence of electrostatic or coordination interactions between the Cu and SiC surface. Therefore, we performed a simple oxygen modification on the surface of SiC to increase the loading of copper atoms. The experimental results revealed that under classical reaction conditions, $CuCl_2$@O−SiC possessed the optimal catalytic activity, producing an isolated yield of up to 75% of only the β,γ-addition product (**2a**) (Table S1, entry 12). The selectivity control of SiO_2 or g-C_3N_4 on the reaction was minimal (Table S1, entries 13 and 14). The replacement of $CuCl_2$ with CuCl diminished both the reaction yield and the selectivity (Table S1, entry 15). Furthermore, when activated charcoal was employed as a carrier, the anticipated outcomes were not obtained (Table S1, entry 16), indicating that the interaction between the metal support and SiC was the primary factor influencing the regulation of the reaction selectivity.

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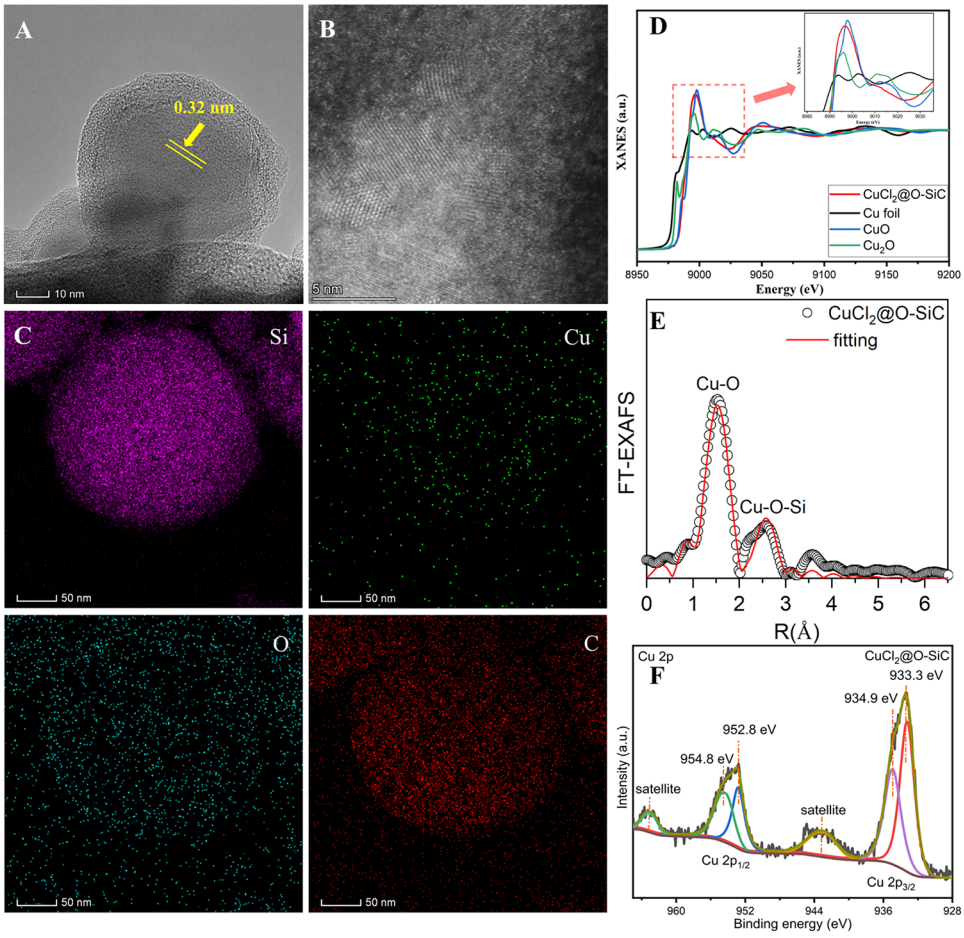


Figure 2. Structural characterization and spectroscopic analysis of $CuCl_2$@O−SiC. (A) HR−TEM image of the $CuCl_2$@O−SiC. (B) AC HAADF-STEM image of the $CuCl_2$@O−SiC. The white particles are Cu SACs. (C) EDS mapping of the $CuCl_2$@O−SiC. (D) Cu K-edge normalized XANES spectra. (E) FT of the k^3-weighted EXAFS spectra. (F) Cu 2*p* XPS spectra of the $CuCl_2$@O−SiC.

Although the catalyst's effectiveness has been demonstrated, $CuCl_2$@O−SiC must be comprehensively characterized to evaluate its catalytic activity. ICP−OES revealed that in $CuCl_2$@O−SiC, the copper loading was only 0.449 wt % (Figure S1). High-resolution transmission electron microscopy (HR−TEM) revealed that no clusters formed during the $CuCl_2$@O−SiC synthesis (Figure 2A). The lattice spacing, measured using HR−TEM, was approximately 0.32 nm. Moreover, HR−TEM clearly revealed that the SiC support possessed lattice fringes. Aberration-corrected high-angle annular dark-field scanning transmission electron microscopy (AC HAADF−STEM) revealed that in $CuCl_2$@O−SiC, copper atoms were uniformly dispersed, the density had increased, and no aggregates had formed (Figure 2B). Energy-dispersive spectrometry mapping (EDS mapping) revealed that in $CuCl_2$@O−SiC, Cu, Si, and C were uniformly dispersed and that the low Cu content was highly dispersed, consistent with the ICP−OES results (Figure 2C). Collectively, these results indicate the absence of copper clusters in the catalyst. The X-ray diffraction (XRD) patterns of both the $CuCl_2$@O−SiC and SiC support were nearly identical and did not exhibit any obvious peaks corresponding to copper nanoparticles, further indicating the absence of copper particles in the catalyst (Figure S2). To gain deep insights into the catalyst's surface oxidation structure, the catalyst was characterized using Fourier transform infrared spectroscopy (FT-IR) and temperature-programmed desorption (TPD). The FT-IR spectrum exhibited distinct broad absorption peaks at 3411 and 1589 cm^{-1}, assigned to hydroxyl group vibrations, suggesting the formation of abundant hydroxyl functionalities on the oxidized catalyst surface (Figure S3). The NH_3-TPD spectrum of the O−SiC exhibited a distinct desorption peak at 537 °C, which was attributed to the acidic sites of the surface hydroxyl groups. Notably, following metal loading, different desorption peaks, probably associated with the interaction between the metallic copper and oxygen atoms of the support, emerged at 160 and 351 °C (Figure S4).[56]

To further investigate the microstructural coordination of the catalyst, we conducted X-ray absorption near-edge structure (XANES) spectroscopy, revealing that $CuCl_2$@O−SiC contained both Cu^{2+} and Cu^{+} species, as confirmed by comparison with standard copper oxide (CuO) and Cu_2O reference spectra. In addition to electronic states, the Cu microstructural coordination environment in the SAC was investigated using Fourier-transformed extended X-ray absorption fine structure (FT-EXAFS) spectroscopy (Figure 2D and Table S2), revealing that in the catalyst, Cu atoms coordinated with four oxygen atoms on average. In *R*-space, the fitting results revealed that Cu−O and Cu−O−Si coordinations were predominant in the first and second shells, respectively (Figure 2E). X-ray photoelectron spectroscopy (XPS) revealed that after the oxygen-based modification of the catalyst surface, the

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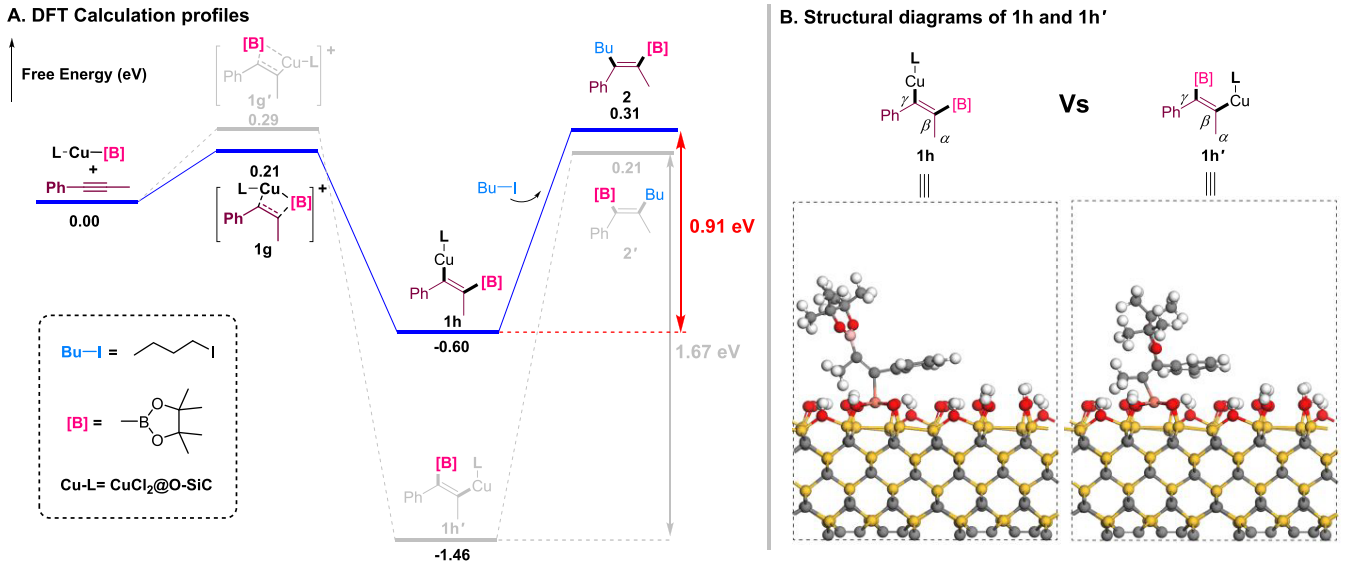


Figure 3. (A) DFT calculations for β,γ-addition. (B) Structural diagrams of **1h** and **1h**′.

O-SiC spectrum exhibited Si 2*p* peaks at binding energies of 102.3 and 100.8 eV, indicating that Si 2p orbitals were in distinct chemical environments, corresponding to the silicon atoms in SiO_2 and SiC, respectively. Simultaneously, the O 1s peaks appeared at 533.1 and 532.3 eV, which were assigned to abundant hydroxyl groups and SiO_2, respectively. Notably, when copper was loading into the catalyst, a different O 1s peak appeared at 531.2 eV, and the Si 2p binding energy shifted because Cu−O bond formation disrupted the original Si−O bond equilibrium (Figure S5). Additionally, in the Cu 2*p* XPS spectrum of $CuCl_2$@O−SiC, peaks at 933.3 and 952.8 eV correspond to the Cu $2p_{3/2}$ peaks of Cu^+, while those at 934.9 and 954.8 eV represented Cu^{2+} characteristic peaks (Figure 2F), indicating that Cu^{2+} and Cu^+ coexisted on the surface of the chemically modified $CuCl_2$@O−SiC. To elucidate the catalysts' Cu^+ source, we loaded the copper into the unmodified carriers and characterized the same using XPS. The spectra only exhibited peaks corresponding to Cu^{2+} species (Figure S6), confirming electronic interactions between the Cu and Si/O carriers in the oxygen-modified catalysts, which effectively modulated the electron density distribution and valence state of the Cu catalytic centers.

To further corroborate the experimental measurements, first-principles density functional theory (DFT) calculations were performed to optimize the theoretical steady-state structures of the $CuCl_2$@O−SiC SACs. Before the reaction, structural optimization revealed that copper atoms adsorbed on oxygen atoms were more stable than those adsorbed on silicon atoms, producing a Cu−O_4 Cu coordination environment (Figure S7),[57] which is highly consistent with the EXAFS fitting.

Studies of the reaction mechanism have revealed that the generation of internal alkyne intermediates from allenes in situ is a crucial first step in controlling the reaction selectivity (Figure S8). Because allenes possess a multitude of potential active sites, extensive computational modeling was conducted to investigate the energy differences among the various reaction intermediates. This study aims to reveal the specific role of the catalyst in promoting the generation of internal alkynes in situ. When copper is bound to the carbon atoms at the β- and γ-positions of allenes, the formed structure possessed the most stable energy, facilitating the formation of internal alkynes (Figure S9). Calculations revealed that the energy required for this transformation was 0.59 eV (Figure S10). The plausibility of the proposed theory was experimentally confirmed. The crude product was reacted with allenes for 1 h under standard conditions and then characterized using ^{13}C NMR spectroscopy, which displayed diagnostic peaks for the alkyne moiety, including a methyl carbon atomic signal at δ 4.04 and *sp*-hybridized carbon atomic resonances at δ 85.58 and 79.46 (Figure S11). The signal coincided with the chemical shift of the alkyne intermediate, providing direct evidence of its in situ generation. Furthermore, the binding of copper catalysts to boron plays a key role in achieving high selectivity for internal alkynes, which is also one of the important factors that enables the reaction to proceed smoothly. Because reaction selectivity control depended entirely on the boronization of the Cu-metal catalytic center with alkynes, the Cu−Bpin reactive species produced different alkyne intermediates because of the different attack directions. Using the basic computational model of Cu SAC, we systematically investigated the energy differences of various reaction intermediates during copper borylation and in the presence of an iodoalkane, revealing the regulatory role of the catalyst in reaction selectivity. Theoretical calculations revealed that when the copper active center coordinated to the γ-position of the alkyne and reacted with the iodoalkane, the activation energy barrier for the formation of the target product was only 0.91 eV. Conversely, when copper bound to the β-position of the alkyne and reacted with the iodoalkane, a substantially higher activation energy barrier (1.67 eV) hindered target product formation, indicating that the catalyst controlled the reaction selectivity. The theoretical calculations and experimental measurements agreed, confirming that the catalyst system's exclusively feasible pathway only generated β,γ-addition products (Figures 3 and S12). In parallel, we investigated the kinetics of the reaction between allenes (**1a**) and iodohydrocarbons and found that the reaction rate depended on the concentration of either the catalyst and the electrophile (1-iodobutane), indicating that the reaction followed first-order kinetics (Figures S13 and S14). As the reaction rate law was consistent with the

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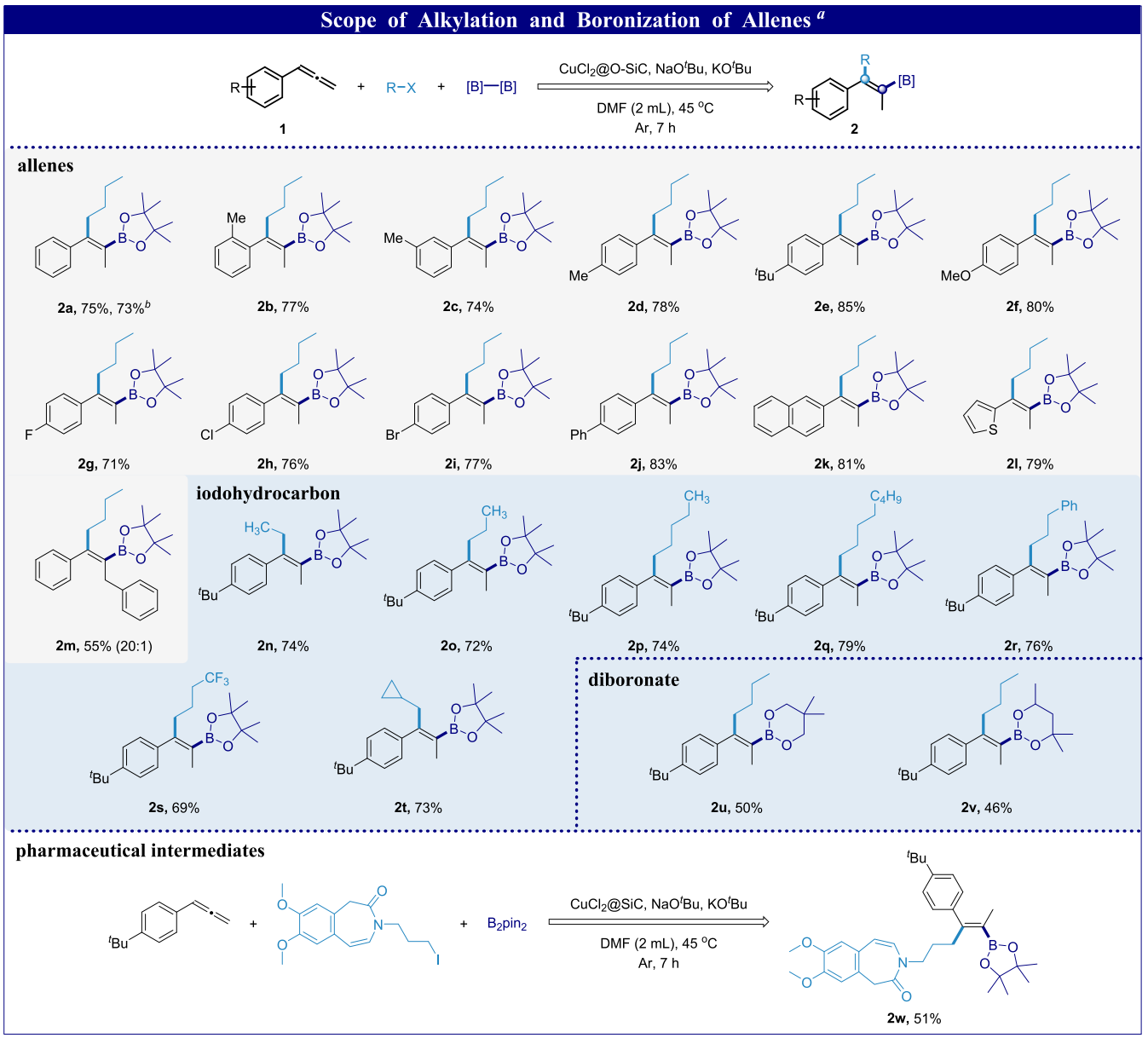


Figure 4. Scope of copper-catalyzed highly selective synthesis, alkylation, and boronization of allenes. [a]Reaction conditions: **1a** (0.2 mmol), iodohydrocarbons (0.6 mmol), B source (0.26 mmol), $CuCl_2$@O−SiC (10 mg), NaO^tBu (0.3 mmol), and KO^tBu (0.3 mmol) in DMF (2 mL) at 45 °C for 7 h. Yields of the isolated products. [b]Yield of **2a** isolated from a large-scale reaction.

mechanism proposed based on the DFT calculation, these results together support the proposed mechanism.

EXPERIMENTAL SECTION

We began by investigating the butylboronization of allenes (**1a**) with 1-iodobutane and bis(pinacolato)diboron under different basic (Table S3) and temperature (Table S4) conditions. The results showed that the optimal conditions were **1a** (0.2 mmol), iodohydrocarbons (0.6 mmol), B source (0.26 mmol), $CuCl_2$@O−SiC (10 mg), NaO^tBu (0.3 mmol), and KO^tBu (0.3 mmol) in *N*,*N*-dimethylformamide (DMF) (2 mL) at 45 °C for 7 h.

Under the optimized conditions, we investigated the range of allenes suitable for the reaction. Allene substrates containing electron-donating groups (Figure 4, **2b**−**2f**, **2j**), electron-absorbing groups (Figure 4, **2g**−**2i**), naphthalene rings (Figure 4, **2k**), and heterocyclic rings (Figure 4, **2l**) produced excellent yields of the target products. Moreover, when these substituents were at different positions in the benzene ring, they only minimally influenced the reaction yields (Figure 4, **2b**−**2d**). Notably, when internal allenes were used, both the reaction selectivity and yield slightly decreased, which may be related to allenes' structure (Figure 4, **2m**). To further extend the applicability of the catalyst, we chose high-yield **1e** as the reaction feedstock and attempted to use iodocarbons (Figure 4, **2n**−**2t**) and different B sources (Figure 4, **2u**−**2v**) for the reaction. The results revealed that the catalyst functioned well with these substrates. In addition, we attempted the reaction using a drug molecule as an organic compound fragment and produced compound **2w**. Selected examples of unsuccessful attempts are compiled in Table S5.

To systematically evaluate the catalytic efficiency and industrial application potential of this catalyst, we conducted a large-scale reaction, revealing that both the product yield and

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the selectivity remained virtually unchanged. Similarly, to assess the catalyst reusability, the heterogeneous catalyst was recovered by washing and filtration after each run. Remarkably, the catalyst could be effectively reused for at least six consecutive cycles without losing any of its catalytic efficiency (Figure S15). After the reaction, the catalysts were analyzed using various characterization means, such as HR-TEM (Figure S16A), STEM (Figure S16B), EDS mapping (Figure S16C−F), AC HAADF-STEM (Figure S17), XPS (Figure S18) and EXAFS (Figure S19), revealing that the catalysts' morphology, composition, and structure negligibly changed, thereby substantiating the catalysts' stability under the prevailing reaction conditions.

CONCLUSIONS

Overall, the synthesis of complex multisubstituted olefins presents formidable challenges, particularly in accelerating reactions, maintaining catalyst stability, and fine-tuning the reaction stereoselectivity. To overcome these challenges, we synthesized a SiC-supported copper SAC ($CuCl_2$@O−SiC) containing increased Cu loadings via oxygen-based surface modification and used the catalyst in the multicomponent boronization of special β,γ-addition-associated branched olefins. These findings should advance research on the synthesis of complex multisubstituted olefins and provide more possibilities for advances and breakthroughs in synthetic chemistry.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.5c03915.

Experiment procedures and ^{1}H and ^{13}C NMR spectra of products (PDF)

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Author Contributions

‖S.R. and S.L. contributed equally to this work. Conceptualization: H.-T.T. and Y.-M.P. Sample preparation, characterization, and theoretical calculation: Z.R., Y.L., and Q.H. Project guidance: M.-R.W., F.-H.C., and W.-H.L. Analyzing data and writing: S.R. and S.L. All authors have approved the final version of the article before submission.

Funding

This work was supported by the National Natural Science Foundation of China (22201049, 22471046), Guangxi Natural Science Foundation of China (2021GXNSFFA220005), the Ba-Gui Youth Top-notch Talents Project of Guangxi, and the National High-Level Personnel of Special Support Program for Young Top-notch Talents (ninth batch) for financial support.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We thank the 1W1B beamlines of Beijing Synchrotron Radiation Facility (BSRF) measurement for providing beam time to support this work. The authors extend their gratitude to Mr. Liao Jian-Liang from Shiyanjia Lab (www.shiyanjia.com) for providing invaluable assistance with the XAFS analysis.

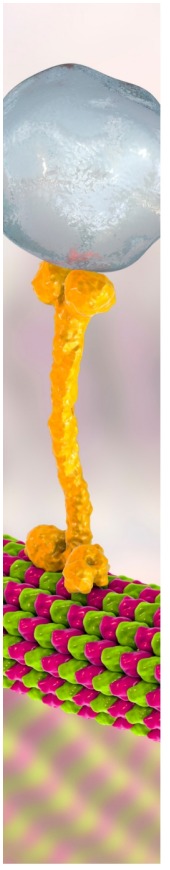
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https://doi.org/10.1021/acscatal.5c03915
ACS Catal. 2025, 15, 15902−15909